

IV FLUIDS — CLINICAL-BASED MCQs

1.

A 55-year-old man with vomiting and dehydration is started on IV fluids. Which fluid is most appropriate for initial rehydration?

- A. 0.45% Normal saline
- B. 0.9% Normal saline
- C. 5% Dextrose
- D. Ringer's lactate

Answer: B

Rationale: Normal saline (0.9%) is isotonic and effective for expanding extracellular volume in dehydration.

2.

A nurse is administering IV fluid to a patient with burns. Which fluid is commonly used for burn resuscitation?

- A. Ringer's lactate
- B. 5% Dextrose
- C. 0.9% Normal saline
- D. 10% Dextrose

Answer: A

Rationale: Ringer's lactate closely mimics plasma electrolytes and is ideal for burn patients to correct fluid loss and acidosis.

3.

A patient with diabetic ketoacidosis (DKA) is receiving IV fluids. The nurse expects to start with:

- A. 5% Dextrose
- B. Ringer's lactate
- C. 0.9% Normal saline
- D. 0.45% Normal saline

Answer: C

Rationale: Isotonic saline is given first to restore circulating volume before switching to hypotonic or dextrose fluids.

4.

Which IV fluid is hypotonic?

- A. 0.45% Normal saline
- B. 5% Dextrose in water
- C. 0.9% Normal saline
- D. Ringer's lactate

Answer: A

Rationale: 0.45% saline has a lower osmolarity than plasma, thus hypotonic.

5.

A patient on IV 0.45% NaCl develops confusion and headache. What should the nurse suspect?

- A. Hypernatremia

- B. Hyponatremia
- C. Hyperkalemia
- D. Hypocalcemia

Answer: B

Rationale: Hypotonic fluids can dilute plasma sodium, causing hyponatremia and cerebral edema.

6.

Which IV fluid should be avoided in patients with liver failure?

- A. Ringer's lactate
- B. Normal saline
- C. 5% Dextrose
- D. 10% Dextrose

Answer: A

Rationale: Ringer's lactate contains lactate, which the liver converts to bicarbonate; impaired metabolism can worsen acidosis.

7.

A patient receiving IV fluids develops puffy eyelids, dyspnea, and crackles. What should the nurse do first?

- A. Increase IV rate
- B. Stop the infusion
- C. Change fluid type
- D. Raise the patient's legs

Answer: B

Rationale: These are signs of fluid overload—infusion should be stopped immediately.

8.

Which of the following fluids provides both calories and free water?

- A. 5% Dextrose in water
- B. 0.9% Normal saline
- C. Ringer's lactate
- D. 3% NaCl

Answer: A

Rationale: 5% dextrose supplies calories (glucose) and free water after metabolism.

9.

A nurse is caring for a patient with cerebral edema. Which fluid should be administered?

- A. 3% NaCl
- B. 0.45% NaCl
- C. 5% Dextrose
- D. Ringer's lactate

Answer: A

Rationale: Hypertonic saline draws water out of swollen brain cells, reducing cerebral edema.

10.

A postoperative patient requires IV fluid for maintenance. Which is most appropriate?

- A. 5% Dextrose in 0.45% NaCl with KCl
- B. 3% NaCl
- C. Ringer's lactate
- D. 0.9% NaCl only

Answer: A

Rationale: This combination provides water, glucose, and electrolytes for daily maintenance.

11.

What is the primary nursing responsibility during blood transfusion with IV fluid?

- A. Use D5W to dilute blood
- B. Administer blood with normal saline only
- C. Use Ringer's lactate
- D. Flush line with hypotonic fluid

Answer: B

Rationale: Only 0.9% normal saline is compatible with blood products; other fluids cause hemolysis.

12.

A patient on IV 3% NaCl should be closely monitored for:

- A. Hypokalemia
- B. Hypernatremia
- C. Hypoglycemia
- D. Hypocalcemia

Answer: B

Rationale: Hypertonic saline can raise serum sodium levels excessively.

13.

A nurse is preparing to give IV potassium chloride (KCl). What is the correct practice?

- A. Give undiluted KCl IV push
- B. Dilute KCl in IV fluid and infuse slowly
- C. Mix KCl with blood transfusion
- D. Add KCl to a hanging IV bag

Answer: B

Rationale: KCl should always be diluted and infused slowly to prevent cardiac arrest.

14.

A patient with congestive heart failure (CHF) requires IV therapy. Which fluid is preferred?

- A. 0.9% NaCl
- B. Ringer's lactate
- C. 0.45% NaCl
- D. Restrict fluids

Answer: D

Rationale: CHF patients retain fluid; fluid restriction prevents volume overload.

15.

Which IV fluid can cause hyperchloremic metabolic acidosis if given in excess?

- A. 0.9% NaCl
- B. Ringer's lactate
- C. 5% Dextrose
- D. 0.45% NaCl

Answer: A

Rationale: High chloride content in normal saline may cause metabolic acidosis.

16.

A nurse notes redness and swelling at an IV site. What is the most likely complication?

- A. Air embolism
- B. Phlebitis
- C. Infiltration
- D. Fluid overload

Answer: B

Rationale: Phlebitis presents as pain, redness, and warmth along the vein.

17.

Which fluid is safest for a patient with hypernatremia?

- A. 5% Dextrose in water
- B. 0.9% NaCl
- C. 3% NaCl
- D. Ringer's lactate

Answer: A

Rationale: D5W provides free water to correct hypernatremia.

18.

A patient with severe diarrhea is losing both water and electrolytes. The best fluid replacement is:

- A. Ringer's lactate
- B. 5% Dextrose
- C. 0.45% NaCl
- D. 10% Dextrose

Answer: A

Rationale: Ringer's lactate restores both fluids and electrolytes effectively.

19.

In a patient with renal failure, which IV fluid should be avoided?

- A. Ringer's lactate
- B. D5W
- C. 0.9% NaCl
- D. 0.45% NaCl

Answer: A

Rationale: Ringer's lactate contains potassium, which can worsen hyperkalemia in renal failure.

20.

Which IV solution is best used to replace fluid loss from nasogastric suctioning?

- A. 0.9% NaCl
- B. 0.45% NaCl
- C. Ringer's lactate
- D. D5W

Answer: A

Rationale: Gastric losses are rich in sodium and chloride; normal saline replaces these effectively.

21.

A patient receiving IV fluids complains of pain at the site with pale, cool skin. What is the likely cause?

- A. Infiltration
- B. Phlebitis
- C. Infection
- D. Air embolism

Answer: A

Rationale: Infiltration occurs when IV fluid leaks into surrounding tissue, causing coolness and swelling.

22.

What is the main indication for using colloid solutions like albumin?

- A. Dehydration
- B. Hypovolemia due to blood loss
- C. Hyponatremia
- D. Maintenance therapy

Answer: B

Rationale: Colloids expand plasma volume by drawing fluid from interstitial to intravascular space.

23.

Which fluid is hypertonic?

- A. 5% Dextrose in 0.9% NaCl
- B. 0.9% NaCl
- C. 0.45% NaCl
- D. Ringer's lactate

Answer: A

Rationale: D5NS is hypertonic and expands extracellular volume.

24.

In a hypoglycemic patient unable to take oral glucose, which IV fluid is appropriate?

- A. 5% Dextrose
- B. 10% Dextrose
- C. 0.9% NaCl
- D. Ringer's lactate

Answer: B

Rationale: D10W provides a higher concentration of glucose to correct hypoglycemia.

25.

A nurse accidentally administers IV fluid too rapidly. Which complication may occur?

- A. Hypovolemia
- B. Fluid overload
- C. Dehydration
- D. Hypotension

Answer: B

Rationale: Rapid infusion causes circulatory overload, especially in elderly or cardiac patients.

26.

Which IV fluid is used for patients with metabolic alkalosis?

- A. 0.9% NaCl
- B. Ringer's lactate
- C. 5% Dextrose
- D. 3% NaCl

Answer: A

Rationale: Normal saline provides chloride to correct metabolic alkalosis.

27.

A patient on total parenteral nutrition (TPN) should have IV access via:

- A. Peripheral vein
- B. Central venous catheter
- C. Subcutaneous line
- D. Arterial line

Answer: B

Rationale: TPN is hyperosmolar and requires central venous access to prevent vein irritation.

28.

A nurse notices that IV infusion has stopped. What is the first step?

- A. Increase the flow rate
- B. Check for kinks or occlusion in tubing
- C. Reinsert the IV cannula
- D. Flush the line with heparin

Answer: B

Rationale: Always assess the system for mechanical causes before intervening.

29.

Which of the following best describes isotonic solutions?

- A. Draw water into cells
- B. Cause fluid shift into ECF
- C. Have same osmolarity as plasma
- D. Expand intracellular fluid

Answer: C

Rationale: Isotonic fluids maintain equal osmolarity with plasma, preventing fluid shift.

30.

A patient on IV therapy for 12 hours has decreased urine output and high blood pressure. What should the nurse suspect?

- A. Dehydration
- B. Fluid overload
- C. Hypovolemia
- D. Hypokalemia

Answer: B

Rationale: Retained fluid increases blood volume and blood pressure with low urine output.